Dr. Sadaqat ur Rehman

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Nationality: British (UK)

Short Biography

Sadaqat ur Rehman is an Adjunct Professor in Artificial Intelligence at the London School of Innovation, London, UK. He is a highly experienced AI scientist with more than 12 years of teaching and research experience in prestigious institutes. He has taught and researched at renowned institutions such as University of Aberdeen, UK, University of Salford, UK, Tsinghua University, China and ETH Zurich, Switzerland, among others. He has also served as a supervisor for several PhD, MSc, undergraduate students and one Post-doc candidate, providing guidance and support for their dissertations.

He has published in renowned journals, including IEEE Transactions, ACM Transactions, Elsevier, and Springer, with more than 95 research papers and over 3,400 citations, achieving an H-index of 30. He has also been named among the world's top 5% scientists in the SCI Ranking 2025. In 2021, he received the UK's Exceptional Talent endorsement as an 'Emerging Leader' for his pioneering work in the field of AI. To date, he has secured research and funding grants of more than £150K in the UK as a Principal Investigator (PI) and Co-Investigator (Co-I). In recognition of his academic excellence, he was awarded a Gold Medal for his outstanding performance in his MS degree.

Education

University of Salford, Manchester, UK

FHEA - Fellowship in Higher Education Academy, UK, Nov 2023

UK Certified Teacher at Higher Education

Tsinghua University, Beijing, China (THE-2026 World Ranking: **12th**)

Ph.D. Information and Communication Engineering – Deep Learning and Computer Vision, Oct 2019

- Thesis Title: *"Investigation of Key Issues in Deep Learning Models for Visual Pattern Classification and Retrieval."*
- Advisory Committee: Profs. Yongfeng Huang and Shanshan Tu.

ETH (Swiss Federal Institute of Technology), Zurich, Switzerland (THE-2025 World Ranking: **6th**)

Exchange Student. Computer Vision Lab, February 2017

- Studied two basic modules of computer vision, *"Image analysis and computer vision, and medical imaging."*
- Research: *"Architecture improvements and Optimization of Deep Learning Models."*

Sarhad University of Science and IT, Pakistan

M.S., Electronics & Communication Engineering – Data Science, 2014

- Ranked first in the Department – **Distinction (Gold Medalist)**
- Research Topic: *"FBVNS: Feature Based Video News Summarization."*
- Advisor: Prof. Rehan Ullah.

University of Engineering and Technology, Peshawar, Pakistan

B.S., Computer Engineering, August 2011

- Final Year Project: *"Comparison Based Analysis of Different Encryption and Cryptographic Techniques using Message Authentication Code in Wireless Sensor Networks."*
- Advisor: Prof. Dr. Khwaja M. Yahya.

Research Funding and Grants

1. Artificial Intelligence based applications for medical education and training, Northern Engineering and Robotics Innovation Centre (NERIC), UK (PI, 2025-2027).
2. AI Literacy for All: A National Framework for Responsible and Inclusive AI Education, PureTech (PI, 2024-26)
3. Efficient and Early-Stage Detection System of Covid-19 based on Respiratory Disorder and Temperature using Deep Learning, PureTech (PI, 2021-2022)
4. Centre for Digital Innovation, University of Salford, UK (Co-PI, 2022).

Research Expertise and Interests

Artificial Intelligence, Machine Learning and Deep Learning, LVM, LLMs, Computer Vision, AI in Healthcare, AI in Education, Image Processing.

Industrial Experience

Artificial Intelligence Engineer Nov 2019 to June 2020

Schlumberger, Beijing Geoscience Center, Beijing, China.

Duties and Responsibilities:

– Implementation of Deep learning/ Machine Learning models for Drilling Dynamics Computation Engines (Well Drilling)

– Delivering lectures to technical audiences for the adaptation of Artificial Intelligence in an oil and gas industry setting.

– Present data and projects clearly and confidently to both foreigners and locals, adapting style and content to the level of knowledge and understanding of audiences

– Communication with vendors and controlling offices regarding different issues

Teaching/ Research Experience

Adjunct Professor in Computer Science March 2025 to date

Department of Computer Science.
Illinois Institute of Technology, Illinois, USA

Teaching Modules:

Data Structures and Algorithms
Operating Systems

Adjunct Professor in AI February 2024 to date

London School of Innovation, London, UK.

Responsibilities:

– Teaching, Research and projects supervision.

Professor (Assistant) in Artificial Intelligence July 2022 to date
Program Leader DipHE Digital and Software Technology

Department of Computing Sciences and Software Engineering, University of Salford, Manchester, UK.

Responsibilities:
– Teaching courses at undergraduate and postgraduate level.
– Communication with University of Salford affiliated colleges in Bahrain and Malaysia.
– Curriculum design at graduate and post-graduate level.
– Collaborating with local and international researchers on AI related projects.
– Generating ideas and writing research proposals for national and international funding calls.
– Supervision of BS, MS and PhD students.
– Co-Pi of Centre for Digital Innovation. Providing supervision/consultancy on AI related projects.
– Program Leader of DipHE Digital and Software Technology.

Teaching Modules :
– Computer Architecture (UG)
– Deep Learning (UG)
– Computer Vision (PG)
– Artificial Intelligence and Data Mining (UG)
– Computer Programming (UG)
– Data Structures and Algorithms (UG)

Teaching Fellow January 2022 to June 2022

School of Natural and Computing Sciences, University of Aberdeen, Aberdeen, UK.

Teaching Modules :
– Computer Architecture (UG)
– Secure Software Design and Development (PG)
– Computer Vision and Image Processing (PG)

Assistant Professor July 2020 to December 2021

Department of Computer Science, Namal Institute - Associated Institute of the University of Bradford, UK.

Responsibilities:
– Teaching courses at undergraduate and postgraduate level.
– Supervision of undergraduate final year projects and interns.
– Curriculum design and laboratory setup.
– Student support and other day to day administrative and managerial activities.
– Collaborating with local and international researchers on AI related projects.
– Generating ideas and writing research proposals for national and international funding calls.

Teaching Modules :
– Introduction to Computer Programming
– Computer Vision
– Image and Signal Processing
– Object Oriented Programming (OOP)
– Discrete Mathematics
– Deep Learning & Neural Networks

Research and Teaching Assistant Sept 2015 to June 2019
School of Information and Communication Engineering, Tsinghua University, Beijing, China.

Responsibilities:
– Taught, research, supervised and demonstrated in Next Generation Networking laboratory
– Assessed students' work, providing written and oral feedback on their performance
– Supervised undergraduate and postgraduate final year projects and interns
– Invigilated in examinations and participating in open days.

Courses Taught:
– Computer Programming
– Computer Architecture
– Data Communication
– Embedded Systems
– Artificial Intelligence
– Data Science
– Computer Vision and Pattern Recognition
– Machine Learning
– Research Methodology

Visiting Researcher Sept 2016 to Feb 2017
Computer Vision Lab (CVL), ETH Zurich, Zurich, Switzerland.
– Image Analysis and Computer Vision
– Medical Image Processing
– Medical Image Registration for MIR Data
– Lab Director: Prof Luc Van Gool

Lecturer Jan 2012 to Aug 2015
Department of Electrical Engineering, Sarhad University of Science and IT, Pakistan.

Responsibilities:
– Taught and Designed undergraduate curriculum in the subject area of Computer Engineering.
– Prepared entrance tests for admission, student assessment, and selection.
– Organized conferences, seminars, and events at SUIT.
– Collaborated with local and international researchers on AI-related projects.
– Design of the curriculum and laboratory setup.
– Student support and other daily administrative and managerial activities.

Quality Enhancement Cell (QEC) Member July 2020 to Dec 2021
Jan 2012 to Aug 2015
Department of Computer Science, Namal Institute, Pakistan.
Department of Electrical Engineering, Sarhad University of Science and IT, Pakistan.

Responsibilities:
– Responsible for promoting public confidence that the quality and standards of the award of degrees are enhanced and safeguarded.
– Responsible for the review of quality standards and the quality of teaching and learning in each subject area.
– Responsible for the review of academic affiliations with other institutions in

terms of effective management of standards and quality of programs.
– Responsible to develop qualifications framework by setting out the attributes and abilities that can be expected from the holder of a qualification, i.e. Bachelors, Bachelor with Honors, Master's, M. Phil., Doctoral.
– Responsible to develop program specifications. These are standard set of information clarifying what knowledge, understanding, skills and other attributes a student will have developed on successfully completing a specific program.
– Responsible to develop quality assurance processes and methods of evaluation to affirm that the quality of provision and the standard of awards are being maintained and to foster curriculum, subject and staff development, together with research and other scholarly activities.
– Responsible to ensure that the university's quality assurance procedures are designed to fit in with the arrangements in place nationally for maintaining and improving the quality of Higher Education.

Awards

- **Global Talent - Exceptional Promise in AI** Endorsed by the Royal Academy of Engineering, UK.
- Nominated for Best Early Career Research Award, University of Salford, UK, 2024.
- World top 5% Scientists (Sci Ranking 2025).
- **Best Paper Presentation Award** at IEEE International Conference of Online Analysis and Computing Science (ICOACS 2016), China.
- **Fully funded Sponsorship from the American Association for the Advancement of Science (AAAS)** for Conference participation in Barcelona Spain July 15-17, 2015.
- **Outstanding Contributions in Reviewing Award** from "The Editors of Journal of Neurocomputing" (Impact Factor 4.072).
- **Outstanding Student Feedback Award** from Sarhad University of Science and IT, Pakistan, 2015.
- **Awarded PhD Studentship (worth** $\$, 10, 700$**)**, School of Information and Communication Engineering, Tsinghua University, Beijing, China, 2015 – 2019.
- **Fully Funded Master of Science (MS) Scholarship (worth** $\$8, 000$**)**, Government of Pakistan, 2012 - 2014.
- **First Position (Certificate of Highest Distinction, Gold Medal Award)**, Department of Electrical Engineering, Sarhad University of Science and IT, Pakistan, 2014.

Skills

Programming Languages

Python, Java, MATLAB and C/C++.

Hardware and Simulation Tools

MATLAB Simulink, Proteous, Pspice, Mathtype, Latex, Multisim, FPGA, Digital Oscilloscope, and Micro-controller and Microprocessor Kit.

Associate Editor of Nature Scientific Report Journal.
Acting as a Editorial Board Member for Chaos Theory and Applications
Soft Guest Editor of Computing Approaches to Continuous Software Engineering

Acting as invited Peer Reviewer for (including but not limited to the following high-impact journals:)

- IEEE Transactions on Circuits and Systems for Video Technology (IEEE)
- IEEE Transactions on Consumer Electronics
- IEEE Transactions on Big Data
- ACM Transactions on Sensor Networks
- IEEE/ACM Transactions on Audio, Speech and Language Processing
- IEEE Transactions on Reliability
- IEEE Access (IEEE)
- IET Image Processing
- PlosOne
- Scientific Reports (Nature)
- Soft Computing (Springer)
- Signal Processing (Elsevier)
- Neurocomputing (Elsevier)
- Entropy (MDPI)
- Technical Program Committee Member of International Conference on Big Data and IoT, Toronto, Canada, 2022
- Technical Program Committee Chair ICICSE 2024, Beijing, China
- Reviewer of ECCV 2024
- Technical Program Committee Member of CSE 2018, 2019, 2020
- Technical Program Committee Member of SMART-TECH 2020, 2021
- Technical Program Committee Member of International Conference on Big Data, Machine Learning and IoT (BMLI 2020, 2021, 2022)

Leadership and Management

- External Examiner at NCG Colleges, London, UK
- Program Leader (PL) DipHE Digital and Software Technology, University of Salford, UK
- Communicating with affiliated colleges of the University of Salford, UK in Bahrain and Malaysia.
- Research consultancy to SME's in Centre for Digital Innovation, UK.
- Member of Quality Enhancement Cell (QEC), Sarhad University of Science and IT (2012-2015), Namal Institute (2020-2021) .
- Chair of Pakistan Society (PAKSOC) in Tsinghua University (2017 – 2018)
- In-charge Micro-controllers Lab (Electrical Engineering Department) (2012 – 2015)

Professional Memberships

- Professional member of British Computer Society (BCS)
- Member IEEE, Reg No. 95164916
- Member of ACM
- Member of IEEE young professionals.
- Member of Pakistan Engineering Council (PEC), Reg No. Comp/8521.
- Member of Image processing and Machine Learning Research Group, Sarhad University of Science and IT, Pakistan.
- Member of University of Engineering and Technology Robotics Club, Pakistan.

Supervision & Co-Supervision

- Supervised 28 MSc students and 40 undergraduate students at University

of Salford, UK, University of Aberdeen, UK, Namal University, and Sarhad University, Pakistan
- Supervising 08 undergraduate students at the university of Salford, UK.
- Supervising 06 postgraduate students at the university of Salford, UK.
- Main Supervisor of 3 PhD students and 1 Post-doc candidate at the university of Salford, UK.

PhD Examiner/ Thesis Evaluator

- Mitesh Rajan "Vehicle Incident detection from videos with Deep Learning", University of Salford, UK
- Josh Frawley "Deep Learning-based DDoS detection: A multilayer approach", University of Salford, UK.
- Jawad Ali, "Analyzing and enhancing inter-network communication and modular connectivity in IoT for the development of smart cities, Univeristy of Engineering and Technology, Peshawar, Pakistan
- Sheraz Ahmad, "Optimization of Mobility-Aware Cache Allocation in Wireless Networks" Iqra National University, Peshawar, Pakistan
- Constintine Nhundu, "A novel data-mining context aware techniques selection framework: A case study of selecting algorithm in different dataset", University of Salford, UK

Collaborative Work

– Dr. Jawad Ahmad *(PMU)*
– Prof. Shanshan Tu *(Beijing University of Technology, Beijing, China)*
– Prof. Yongfeng Huang *(Tsinghua University, Beijing, China)*
– Prof. Ting Zhang *(Beijing University of Technology, Beijing, China)*
– Dr. Aziz Shah *(Coventry University, UK)*
– Prof. Wadii Boulila *(Prince Sultan University, KSA)*
– Prof. Bacha Rehman *(Solent University, UK)*
– Dr. Chathura Madhusanka *(University of Essex, UK)*
– Dr. Zubair Shah *(Hammad Bin Khalifa University, Qatar)*
– Prof. Nicola Marchetti *(Trinity College Dublin, Ireland)*
– Prof. Chin-Chen Chang *(Feng Chia University, Taichung City, Taiwan)*
– Prof. Khwaja M. Yahya *(Umm al Qura University, Saudi Arabia)*
– Prof. Obaid Rehman *(Sarhad University of Science and Information Technology, Pakistan)*
– Dr. Muhammad Waqas *(Greenwich University, London, UK)*
– Prof. Salman Ahmed *(University of Engineering and Technology, Peshawar, Pakistan)*
– Prof. Shiyou Yang *(Zhejiang University, Hangzhou, China)*
– Prof. Luc Vangool *(Swiss federal Institute of Technology, Zurich, Switzerland)*
– Prof. Anis Kouba *(Prince Sultan University, Saudi Arabia)*
– Prof. Hasan Al Marzouqi *(Khalifa University, Abu Dhabi, UAE)*

Research Publications

Cumulative Impact Factor (as per JCR, 2026): 400+
Cumulative Google Citation: 3450
Google H-Index: 31
Google I10-Index: 67
Google Scholar Link:
https://scholar.google.com/citations?user=FHHxWXMAAAAJhl=enoi=ao
Research Gate Link:
https://www.researchgate.net/profile/Sadaqat_Rehman/research
Publons: Reviewed 150+ journal papers
Publons Link: https://publons.com/researcher/3046438/sadaqat-ur-rehman/

Journal Articles

1. **Sadaqat ur Rehman**, Shanshan Tu, Muhammad Waqas, Obaid ur Rehman, Yongfeng Huangand and Salman Ahmad, Unsupervised pre-training learning approach for Efficient Convolution Neural Network, *Neurocomputing*, July, 2019 **(Impact factor: 6)**
2. **Sadaqat ur Rehman**, Shanshan Tu, Yongfeng Huang and Obaid ur Rehman, A benchmark Dataset and Learning High Level Semantic Embeddings of Multimedia in Cross-media Search, *IEEE Access*, Page(s): 67176 - 67188, Vol. 06, DOI: 10.1109/ACCESS.2018.2878868, October 2018 **(Impact Factor: 3.745)**
3. **Sadaqat ur Rehman**, Shanshan Tu, Yongfeng Huang and Obaid ur Rehman, Optimization of CNN through Novel Training Strategy for Visual Classification Problems", *Entropy*, 20(4), 290; https://doi.org/10.3390/e20040290 **(IF: 2.73)**
4. **Sadaqat ur Rehman**, shanshan Tu and Yongfeng Huang, CSFL: A Novel unsupervised Convolution Neural Network approach for Visual Pattern Classification, *AI Communications*, Vol. 30, no. 5, pp. 311-324, 2017 **(IF: 1.02)**
5. **Sadaqat ur Rehman**, M. Bilal, Basharat Ahmad, K.M. Yahya and Obaid ur Rehman, Comparison Based Analysis of different Cryptographic and Encryption Techniques using MAC in WSN, *International Journal of Computer Science Issues (IJCSI)*, Vol. 9, Issue 1, No 2, ISSN 1694-0814, January 2012 (Elsevier Indexed).
6. **Sadaqat ur Rehman** and Junaid khan, Linear Quadratic Regulator Controller for magnetic levitation System, *International Journal of Computer Science and Software Engineering*, Vol. 4, Issue 1, Pages: 6-9, January 2015.
7. Amin Ullah, **Sadaqat ur Rehman***, Shanshan Tu, Fawad, A Hybrid Deep CNN Model for Abnormal Arrhythmia Detection based on Cardiac ECG Signal, *Sensors*, January 2021, **(Impact Factor: 3.84)**
8. Shanshan Tu, **Sadaqat ur Rehman***, Muhammad Waqas and Obaid ur Rehman, Optimization Based Training of Evolutionary Convolution Neural Network for Visual Classification Applications, *IET Computer Vision*, March 2020 **(Impact Factor: 1.95)**.
9. Ting Zhang, Haijian Shen, **Sadaqat ur Rehman*** and Zhaoying Li, Two-stage Domain Adaptation for Infrared Ship Target Segmentation, *IEEE Transactions on Geoscience and Remote Sensing*, March, 2024 **(Impact Factor: 7.5)**. **Q1**
10. Zhaoying Liu, Xiang Li, Ting Zhang, Bo Liu, **Sadaqat ur Rehman***, Bacha Rehman, Changming Sun, Saliency Guided Siamese Attention Network for Infrared Ship Target Tracking, *IEEE Transactions on Intelligent Vehicles*, 2024 **(Impact Factor: 14)**.
11. Xiang Li, Ting Zhang, Joying Liu, **Sadaqat ur Rehman***, Saliency Guided Siamese Attention Network for Infrared Ship Target Tracking, *IEEE Transactions on Intelligent Vehicles*, DOI: 10.1109/TIV.2024.3370233, March 2024 **(Impact Factor: 8.20)**.
12. Zhaoying Liu, Yuxiang Zhang, Junran He, Ting Zhang, **Sadaqat ur Rehman***, Mohamad Saraee, Enhancing Infrared Small Target Detection: A Saliency-Guided Multi-Task Learning Approach, *IEEE Transactions on Intelligent Transportation System*, January 2025 **(Impact Factor: 7.9)**.
13. Shanshan Tu, **Sadaqat ur Rehman***, Muhammad Waqas, ModPSO-CNN: An Evalutionary Convolution Neural Network for Visual Classification Problem, *Soft Computing*, September 2020 **(Impact Factor: 3.73)**.

14. Shanshan Tu, Muhammad Waqas, **Sadaqat ur Rehman**, and Talha Mir, Reinforcement Learning Assisted Impersonation Attack Detection in Device-to-Device Communications, *IEEE Transactions on Vehicular Technology*, Vol. 70, Issue 2, pp. 1474-1479, February 2021 **(Impact Factor: 6.23)**
15. Obaid ur Rehman, Shiyou Yang, Shafi Ullah and **Sadaqat ur Rehman**, A Quantum Particle Swarm Optimizer with Enhanced Strategy for Global Optimization of Electromagnetic Devices, *IEEE Transactions on Magnetics*, Vol. 55, Issue 08, Page 1-4, August 2019 **(Impact Factor: 1.84)**.
16. Ting Zhang, Muhammad Waqas, Joying Liu, **Sadaqat Ur Rehman**, and Shanshan Tu, A Fusing Framework of Shortcut Convolutional Neural Networks, *Information Sciences*, August 2021, DOI: 10.1016/j.ins.2021.08.030 **(Impact factor: 8.23)**
17. Jahanzaib Latif, Shijin Zhang, **Sadaqat Ur Rehman***, Azhar Imran, Enhancing surveillance and early warning of infections and antimicrobial resistance using machine learning and deep learning: A systematic review, *Neurocomputing*, 684:133502 DOI: 10.1016/j.neucom.2026.133502, March 2026 **(Impact Factor: 5.50)**.
18. Ting Zhang, Qiyu Yang, Haijian Shen, **Sadaqat Ur Rehman***, DADANet: data-augmented domain adaptation with region-guided pseudo-label correction for infrared ship semantic segmentation, *Multimedia Systems* , 32(4), May 2026
19. Basharat Ahmad, Yamin Li, Zhaoliang Wu, Sadaqat Ur Rehman, Yongfeng Huang, Improved attack detection in IoT and IIoT networks using attention mechanisms in convolutional neural networks, *Expert Systems With Applications*, Volume 296, Part C, 129021, July 2025 **(Impact Factor: 7.50)**.
20. Momina Qureshi, Arbab Masood Ahmad, **Sadaqat ur Rehman***, Deep learning-based forecasting of electricity consumption, *Nature Scientific Reports*, DOI: 10.1038/s41598-024-56602-4, March 2024 **(Impact Factor: 4.60)**.
21. Ting Zhang, Tianyang You, Zhaoying Liu, **Sadaqat ur Rehman***, Yanan Shi, Amr Munshi, Small sample pipeline DR defect detection based on smooth variational autoencoder and enhanced detection head faster RCNN, *Applied Intelligence*, Volume 55, Issue 8, Pages 1-29, 2025
22. Muhammad Hammad, Fakhia Hammad, Muhammad Taha, Shivakumara Palaiahnakote, **Sadaqat ur Rehman**, Mohamad Saraee, A novel data-centric AI approach based on sensitivity and correlation analyses for multi-organ plant disease classification, *Expert Systems with Applications*, Pages 128365
23. Mir, S., Arbab, M.A. & **Sadaqat ur Rehman***. ENSO dataset & comparison of deep learning models for ENSO forecasting. Earth Sci Inform (2024). https://doi.org/10.1007/s12145-024-01295-6
24. Tan Haining, Tao Ye, **Sadaqat ur Rehman***, Obaid ur Rehman, Shanshan Tu, A Novel Routing Optimization Strategy Based on Reinforcement Learning in Perception Layer Networks, *Computer Networks*, Volume 237, December 2023 **(Impact Factor: 5.60)**. **Q1**
25. Jahanzaib Latif, Chuangbai Xiao, Shanshan Tu, **Sadaqat Ur Rehman**, Azhar Imran, Anas Bilal, Implementation and Use of Disease Diagnosis Systems for Electronic Medical Records Based on Machine Learning: A Complete Review, *IEEE ACCESS*, August 2020, DOI: 10.1109/ACCESS.2020.3016782 **(Impact factor: 3.745)**
26. Basharat Ahmad, Yongfeng Huang and **Sadaqat Ur Rehman**, Enhancing the security in IoT and IIoT networks: An intrusion detection scheme leveraging deep transfer learning, *Knowledge-Based System*, September 2024 **(Impact factor: 7.2)**

27. ur Rehman, Zia, Zeeshan Aziz, Usama Khalid, Nauman Ijaz, **Sadaqat Ur Rehman**, and Zain Ijaz. "Artificial intelligence-driven enhanced CBR modeling of sandy soils considering broad grain size variability." Journal of Rock Mechanics and Geotechnical Engineering (2024) **(Impact factor: 9.4)**.

28. Ting Zhang, **Sadaqat ur Rehman***, Bo Liu, Muhammad Saraee, Gas Pipeline Defect Detection based on Improved Deep Learning, *Expert Systems With Application*, December 2024, **(Impact Factor: 7.5)**.

29. Andrew Churcher, Rehmat Ullah, Jawad Ahmad, **Sadaqat Ur Rehman** and Fawad Masood, Attack Classification using Machine Learning in IoT Networks *Sensors*, no. 2, p. 446, January 2021, **(Impact factor: 3.84)**

30. Shanshan Tu, Muhammad Waqas, **Sadaqat Ur Rehman**, Talha Mir and Zahid Haleem, Social Phenomena and Fog Computing Networks: A Novel Perspective for Future Networks *IEEE Transactions on Computational Social Systems* DOI: 10.1109/TCSS.2021.3082022, May 2021 **(Impact factor: 4.74)**

31. Muhammad Waqas, Shanshan Tu, Zahid Haleem, **Sadaqat Ur Rehman**, The Role of Artificial Intelligence and Machine Learning in Wireless Networks Security: Principle, Practice and Challenges, *Artificial Intelligence Review*, https://doi.org/10.1007/s10462-022-10143-2, February 2022, **(Impact factor: 9.58)**

32. Fawad Masood, Wadi Boulila, Jawad Ahmad, **Sadaqat Ur Rehman**, A Novel Image Encryption Scheme Based on Arnold Cat Map, Newton-Leipnik System and Logistic Gaussian Map, *Multimedia Tools and Applications* , March 2022 **(Impact factor: 2.75)**

33. Ting Zhang, Gualin Jiang, Zhaoying Liu and **Sadaqat ur Rehman***, Advanced Integrated Segmentation Approach for Semi-supervised Infrared Ship Target Identification, *Alexandria Engineering Journal* **(Impact Factor: 6.80)**. **Q1**

34. Yin Liang, Gaoxu Xu, **Sadaqat Ur Rehman**, Multi-Scale Attention-Based Deep Neural Network for Brain Disease Diagnosis, *CMC-Computers, Materials & Continua*, DOI: http://dx.doi.org/10.32604/cmc.2022.026999, 72(3):4645-4661, January 2022 **(Impact Factor: 3.86)**.

35. Ting Zhang, Pengyi Liu, Haijian Shen, Zhaoying Liu, **Sadaqat Ur Rehman** and Amr Munshi, Dual domain adaptation for infrared ship segmentation, *Optics & Laser Technology*, Volume 195, March 2026, **(IF: 5.0)**.

36. Chaofan Chen, Ting Zhang, Cong Ma, Zhaoying Liu, **Sadaqat Ur Rehman**, L-DC-DETR: A Lite Distance-Content Aware DETR for Gas Pipeline Defect Detection, *Signal, Image and Video Processing*, Volume 20, Issue 1, Pages 16, January 2026

37. Obaid Ur Rehman, Abid Saeed, Sadaqat Ur Rehman, Optimization of electromagnetic devices using a novel search based quantum particle swarm approach, *COMPEL*, DOI: 10.1108/COMPEL-05-2025-0244, 2026

38. Zia Ur Rehman, Obaid Ur Rehman, Amr Munshi, Sadaqat Ur Rehman, Parameters optimization of photovoltaic systems using modified quantum inspired particle swarm method, *Scientific Reports*, DOI: 10.1038/s41598-026-38620-6, 2026

39. Jialin Wan, Muhammad Waqas, Shanshan Tu, Syed Mudassir Hussain, Ahsan Shah, **Sadaqat Ur Rehman** and Muhammad Hanif, An Efficient Impersonation Attack Detection Method in Fog Computing *CMC-Computers, Materials & Continua*, DOI:10.32604/cmc.2021.016260, January 2021 **(IF: 3.86)**.

40. Bin Zhang, Muhammad Waqas, Shanshan Tu, Syed Mudassir Hussain and **Sadaqat Ur Rehman**, Power Allocation Strategy for Secret Key GenerationMethod in Wireless Communications *CMC-Computers, Materials & Continua*, DOI:10.32604/cmc.2021.016553, April 2021 **(Impact Factor: 3.86)**.
41. Fawad Masood, Maha Driss, Wadii Boulila, Jawad Ahmad, **Sadaqat Ur Rehman**, Sana Ullah Jan, A Lightweight Chaos-Based Medical Image Encryption Scheme Using Random Shuffling and XOR Operations, *Wireless Personal Communications*, https://doi.org/10.1007/s11277-021-08584-z, May 2021. **(Impact Factor: 2.01)**
42. Fawad Masood, Junaid Masood, Lejun Zhang, Sajjad Shaukat Jamal, Wadii Boulila, **Sadaqat Ur Rehman**, A new color image encryption technique using DNA computing and Chaos-based substitution box, *Soft Computing*, https://doi.org/10.1007/s00500-021-06459-w, November 2021. **(IF: 3.73)**
43. Jahanzeb Latif, Shanshan Tu, Chuangbai Xiao, **Sadaqat Ur Rehman**, Digital Forensics Use Case for Glaucoma Detection Using Transfer Learning Based on Deep Convolutional Neural Networks, DOI: 10.1155/2021/4494447, November 2021 *Security and Communication Networks* **(Impact Factor: 1.96)**
44. Wanxia Yang, **Sadaqat Ur Rehman** and Wenhui Que, Identifying Event-specific Opinion Leaders by Local Weighted LeaderRank, *Intelligent Automation & Soft Computing* 26(5):1185-1198, January 2020, **(Impact factor: 3.40)**
45. Guan, Keqing, Shah Nazir, Xianli Kong and **Sadaqat Ur Rehman**, Software Birthmark Usability for Source Code Transformation Using Machine Learning Algorithms. *Scientific Programming*, DOI: 10.1155/2021/5547766, February 2021, **(Impact factor: 0.963)**
46. Obaid ur Rehman, **Sadaqat ur Rehman*** and Shanshan Tu, A Quantum Particle Swarm Optimization Method with Fitness Selection Methodology for Electromagnetic Inverse Problems, *IEEE ACCESS*, Vol. 06, Page(s). 63155 – 63163, DOI:10.1109/ACCESS.2018.2873670, October 2018 **(IF: 3.745)**
47. Zia Ullah, Mohammad Ismail Mohmand, **Sadaqat ur Rehman*** and Shanshan Tu, Emotion Recognition from Occluded Facial Images Using Deep Ensemble Model, *Computers, Materials and Continua*, 73(3):4465-4487, May 2022 **(IF: 3.86)**
48. Shanshan Tu Obaid ur Rehman, **Sadaqat ur Rehman** and Muhammad Waqas, A Novel Quantum Inspired Particle Swarm Optimization Algorithm for Electromagnetic Applications, *IEEE ACCESS*, Vol. 08, Issue 01, pg. 21909-21916, December, 2019 **(Impact Factor: 3.745)**.
49. Shanshan Tu, Muhammad Waqas, **Sadaqat ur Rehman**, Muhammad Hanif, Tracking area list allocation scheme based on overlapping community algorithm, *Computer Networks*, Vol. 173, https://doi.org/10.1016/j.comnet.2020.107182, May 2020 **(Impact Factor: 5.49)**.
50. Shanshan Tu, Meng Liu, Muhammad Waqas and **Sadaqat ur Rehman**, FHC-PCIA: A Physical Cell Identification Allocation Method based on Fuzzy Hierarchical Clustering for Heterogeneous Cellular Network, *IEEE ACCESS*, Vol. 06, page(s): 46976 – 46987, DOI: 10.1109/ACCESS.2018.2867066, August 2018 **(IF: 3.745)**
51. Liu Meng, Shanshan Tu, Muhammad Waqas and **Sadaqat ur Rehman**, The Allocation and Reuse of Physical Cell Identifications Scheme Based on Maximum Degree First Coloring Algorithm, *IEEE System Journal*, April 2019 **(IF: 4.80)**

52. Shanshan Tu, Muhammad Waqas and **Sadaqat ur Rehman**, Security in Fog Computing: A Novel Technique to Tackle an Impersonation Attack, *IEEE ACCESS*, Vol. 06, Page(s): 74993 – 75001, DOI: 10.1109/ACCESS.2018.2884672, December 2018 **(Impact Factor: 3.745)**.
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